



The Compliance Stack Map — LedgerProof x the EU AI Act

Where the open cryptographic provenance protocol plugs into every operative Article, Annex, and conformity-assessment path of Regulation (EU) 2024/1689.

LedgerProof Foundation
Open specification (CC-BY)
Verifiable at [spec.ledgerproof.com](#)

UNACCEPTABLE RISK

Banned outright. Social scoring, real-time biometric ID in public spaces, voice toys encouraging dangerous behaviour.

N/A — systems prohibited

HIGH-RISK

Annex II products + Annex III categories (employment, credit, healthcare, critical infrastructure, law enforcement). Conformity assessment required.

Supporting fit — Articles 8/11/12/13/14/17 evidence substrate

LIMITED RISK

Article 50 transparency obligations. Chatbots, synthetic media, AI-generated text. Applies **2 August 2026**.

Primary fit — 5 of 6 sub-obligations in v1.0

MINIMAL RISK

Spam filters, e-commerce recommendations. Voluntary codes of conduct only.
N/A — no regulatory trigger

ARTICLE	TOPIC	OCPP-AI PLUG-IN	STATUS
CHAPTER II — PROHIBITED PRACTICES			
Art 5	Prohibited AI practices Unacceptable-risk systems banned from the Union market.	Not applicable — OCPP-AI is not for prohibited systems.	N/A
CHAPTER III SECTION 1 — CLASSIFICATION			
Art 6 · (3)	Risk classification + Material Impact Exception Annex I/III tests; downgrade for preparatory-function systems.	Receipts can document classification rationale and Material Impact Exception evidence under Art 6(3).	PLUG-IN
CHAPTER III SECTION 2 — REQUIREMENTS FOR HIGH-RISK AI SYSTEMS (ARTICLES 8–15)			
Art 8	Compliance with the Requirements Meta-obligation: AI system satisfies Articles 9–15.	Receipts are the evidentiary artifact demonstrating Article 11–14 compliance.	PLUG-IN
Art 9	Risk Management System (RMS) Continuous, iterative risk identification + mitigation.	Not addressed — OCPP-AI is a substrate; receipts can feed RMS metrics but do not constitute an RMS.	GAP
Art 10	Data and Data Governance Training/validation data quality controls.	Not addressed — OCPP-AI is downstream of training; client retains data-governance responsibility.	GAP
Art 11	Technical Documentation Pre-market technical file (Annex IV).	Specification + receipt-format documentation forms part of the technical-doc package for any system using the protocol.	PLUG-IN
Art 12	Record-Keeping Automatic recording of events ("logs") over system lifetime.	Primary fit. Cryptographic record-keeping IS the core function. Tamper-evident, Bitcoin-anchored, supervisory-authority verifiable without vendor cooperation.	PRIMARY ✓✓
Art 13	Transparency & Provision of Information to Deployers Instructions for use; capabilities + limitations disclosure.	Foundation publishes complete protocol documentation; receipts enable downstream verifiers (including deployers in chain-of-supply scenarios) to audit compliance.	PLUG-IN
Art 14	Human Oversight Tools and measures enabling human oversight while in use.	HumanReview profile captures reviewer_role, reviewer_organization, review_rationale, review_timestamp — the artifacts that demonstrate Art 14 oversight workflows.	PLUG-IN
Art 15	Accuracy, Robustness & Cybersecurity Resilience against errors and adversarial attempts.	Partial — Ed25519 signatures + anchor immutability address cybersecurity of the evidence layer. Model accuracy/robustness are not in scope.	PARTIAL
CHAPTER III SECTION 3 — PROVIDER OBLIGATIONS			
Art 17	Quality Management System (QMS) Documented compliance strategy, post-market monitoring.	Receipts serve as cryptographic QMS records; per VisionAI+ Annex VI internal-assessment path: "QMS complies with Article 17."	PLUG-IN
CHAPTER III SECTION 5 —			

CONFORMITY ASSESSMENT

Annex VI — Internal Assessment

QMS conforms to Art 17 - technical documentation, post-market monitoring consistent with Art 17(3).

Annex VII — Third-Party Conformity Assessment

Required when: (a) safety of critical systems; or (c) the provider cannot demonstrate all relevant requirements.

The harmonised-standard

CEN-CENELEC JTC 21 reference documents. Self-certification — shortened timeline.

29 FRAMEWORK ADAPTERS

Native plug-in at every major AI framework lines of integration code.

langchain	openai
llamaindex	azure-openai
google-ai	together
xai	deepinfra
cerebras	watermark
aleph-alpha	haystack

GDPR · ARCHITECTURE

Three layers, separated by de-identification at the lowest.

- Schema layer: Personal data structure.
- Receipt store: GDPR Art 17(1) erasure content.
- Anchor layer: 36-byte LPR content.

STANDARDS (ARTICLES 40–42)		
Art 40	Harmonised Standards Presumption of conformity via European harmonised standards.	Strategic lever. OCPP-AI Foundation engaged with CEN-CENELEC JTC 21 to position the protocol as a harmonised standards reference — converts Annex VII third-party assessment into Annex VI self-certification. PLUG-IN
CHAPTER IV — TRANSPARENCY OBLIGATIONS (ARTICLE 50 SUB- OBLIGATIONS)		
Art 50(1)	Direct disclosure of AI interaction Natural person knows they are talking to an AI.	ChatbotSession profile: ai_system_id, deployer_id, disclosure_text_hash, session_start_timestamp. Schema rejects PII at validation. PRIMARY ✓✓
Art 50(2)	Synthetic content marking Machine-readable marking of AI-generated/manipulated content.	GeneratedContent profile: content_hash, perceptual_hash, generation_type, source_content_hash, transparency_marker. PRIMARY ✓✓
Art 50(3)	Emotion recognition & biometric notification Subjects of biometric/affect AI must be informed.	Deferred to v1.1 pending GDPR Art 9 / 22 coordination consultation. Documented gap with explicit consultation pathway. DEFERRED
Art 50(4)	AI-generated text + editorial-review exemption Public-interest text disclosed unless human-reviewed.	HumanReview profile evidences the editorial-review exemption with bounded-vocabulary reviewer fields. PRIMARY ✓✓
Art 50(5)	Composition of 50(1) + 50(2) Cross-cutting disclosure operability.	Addressed by composition of ChatbotSession + GeneratedContent profiles. PLUG-IN
Art 50(6)	Clear & distinguishable manner Disclosure form requirements.	transparency_marker field present across ChatbotSession + GeneratedContent. Default "LPR-EU-AI-ACT-50"; localised values admitted. PRIMARY ✓✓
CHAPTER V — GPAI PROVISIONS (ARTICLES 51–55)		
Art 53+	GPAI provider obligations Transparency, training-data summaries, copyright compliance.	Provider-side receipts at the model output boundary; deployers consuming GPAI outputs gain a verifiable trail for downstream Article 50 obligations. PLUG-IN

INSTITUTIONAL ENGAGEMENT (VERIFIABLE URLS)	
IETF SCITT WG draft	draft-dawkins-scitt-ai-article50-00
EU AI Office consultation	Futurium · filed 2 Jun 2026
DG-CNECT memorandum	delivered 3 Jun 2026
Stibbe Brussels engagement	Annex VI desk-officer sourcing
arXiv preprint	cs.CR · submission in flight
CEN-CENELEC JTC 21	national mirror committee path (DIN-SME)

ANCHOR SUBSTRATE (BITCOIN + EBSI DUAL-ANCHOR)	
Reference substrate	Bitcoin OP_RETURN
Anchor payload	"LPR1" merkle_root_32 (36 bytes)
EU-sovereign alternative	European Blockchain Services Infrastructure (EBSI)
Dual-anchor mode	Recommended for EU-jurisdictional deployers
Signature	Ed25519 over CBOR canonical encoding
Anchor Interface	8 normative properties (I-1 through I-8)

PLUG-IN STATUS LEGEND	
PRIMARY	Core function — the protocol IS the artifact for this Article
PLUG-IN	Direct integration point — OCPP-AI provides evidence sub
PARTIAL	Bounded coverage — documented scope limits.
GAP	Explicit not-addressed — complementary tooling required.
N/A	Not applicable to scope of OCPP-AI.